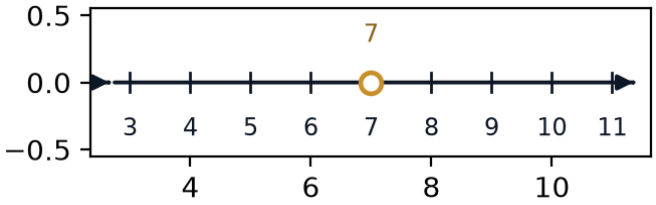
Algebra & Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

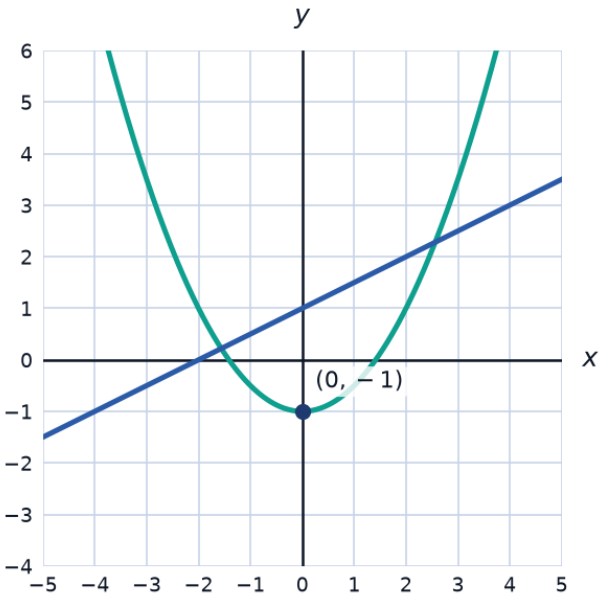
1 [Foundational] • Target: Linear inequalities

Solve $2x - 5 < 9$ and graph the solution set on a number line.



2 [Foundational] • Target: Solving linear equations

Solve $3x - 7 = 11$.



3 [Foundational] • Target: Solving linear equations

Solve $5(x - 2) = 3x + 4$.

4 [Foundational] • Target: Solving linear equations

Solve $\frac{3x - 2}{4} = x - 1$.

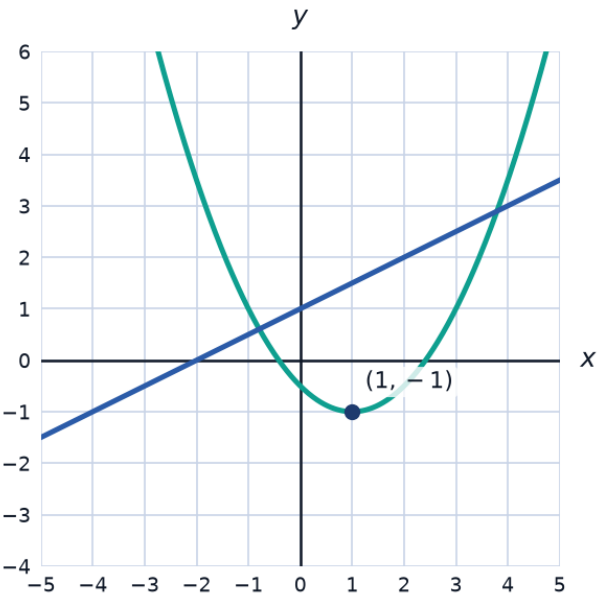
5 [Foundational] • Target: Solving linear equations

Solve $2(x + 3) - 5 = 3(x - 1)$.

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6 [Foundational] • Target: Expanding and simplifying expressions

Expand and simplify $3(2x - 1) - 2(x - 4)$.



7 [Foundational] • Target: Expanding and simplifying expressions

Expand $(x + 5)(x - 2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Expanding and simplifying expressions

Expand $(2x - 3)^2$.

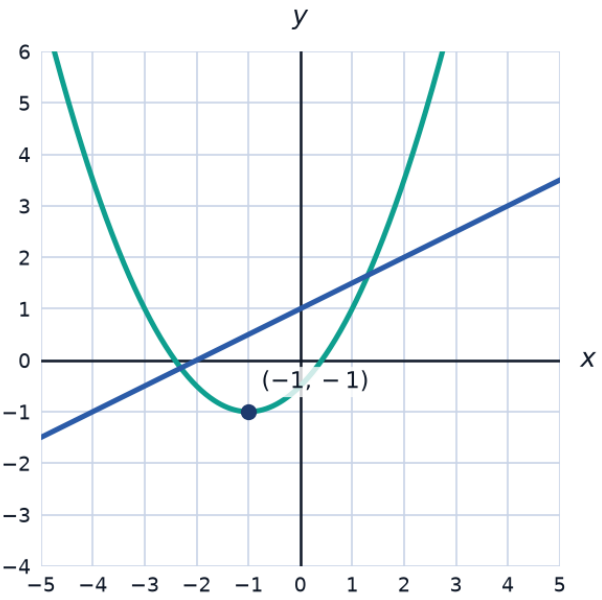
9 [Intermediate] • Target: Expanding and simplifying expressions

Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).

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10 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 + 7x + 12$.



11 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 2x - 15$.

12 [Intermediate] • Target: Factoring quadratic expressions

Factor $2x^2 + 5x - 3$.

13 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 49$ using the difference of squares.

14 [Intermediate] • Target: Solving quadratic equations by factoring

Solve $x^2 - 5x + 6 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 9.01 Statistics & Probability

Solve $x^2 + 2x - 8 = 0$.

16 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 2.01 Functions & Transformations

Solve $2x^2 - 5x - 3 = 0$.

17 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 - 9x + 20 = 0$.

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18 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 4.01 Trigonometry

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

19 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

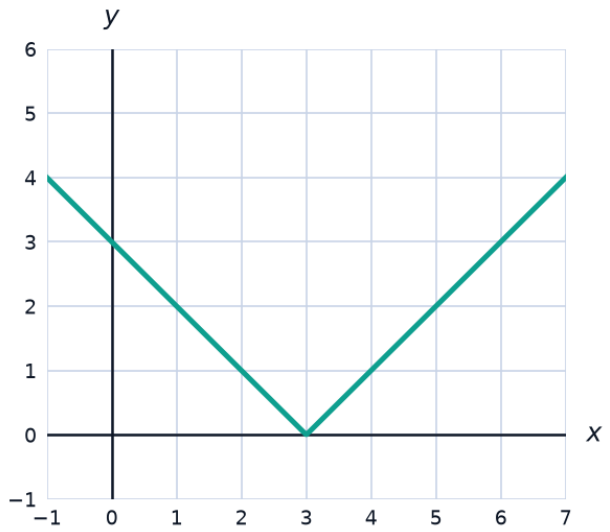
20 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 6.01 Limits & Derivatives

Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Absolute value equations

The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?



a. $x = 1$

b. $x = 3$

c. $x = -3$

d. $x = 0$

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22 [Foundational] • Target: MCQ — Linear inequalities

Solve $3x - 7 < 14$, and identify the graph of its solution set.



a. $x < 7$

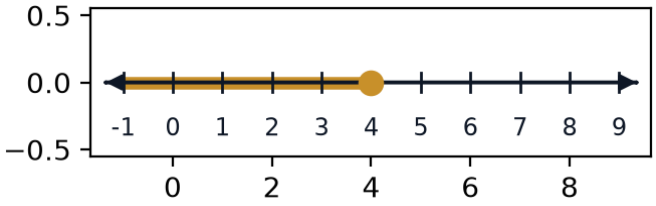
b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

23 [Foundational] • Target: MCQ — Linear inequalities

Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.



a. $x > 4$

b. $x < 4$

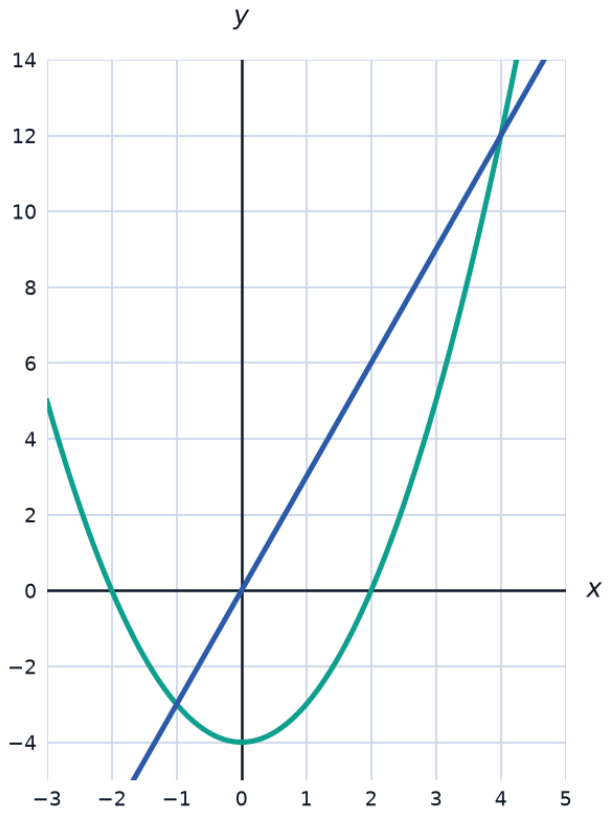
c. $x \leq 4$

d. $x \geq 4$

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24 [Foundational] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?



a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

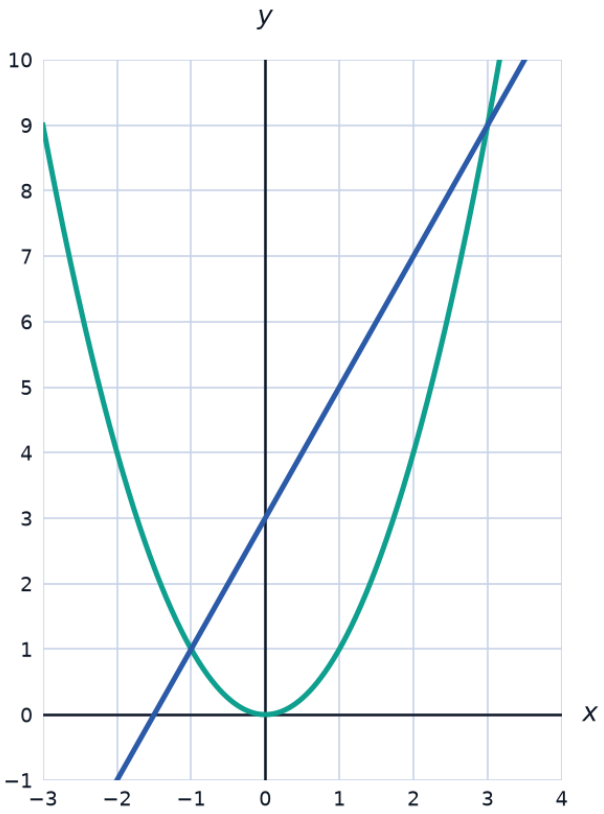
c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

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25 [Foundational] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?



a. $(3, 9)$ and $(1, 1)$

b. $(-1, 1)$ and $(1, 1)$

c. $(-3, 9)$ and $(1, 1)$

d. $(3, 9)$ and $(-1, 1)$

26 [Foundational] • Target: MCQ — Solving linear equations

Solve $4x - 9 = 15$.

a. $x = 6$

b. $x = -6$

c. $x = 24$

d. $x = 1.5$

27 [Foundational] • Target: MCQ — Solving linear equations

Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$

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28 [Foundational] • Target: MCQ — Factoring quadratic expressions

Factor $x^2 + 9x + 14$.

a. $(x + 2)(x + 7)$

b. $(x - 2)(x - 7)$

c. $(x + 1)(x + 14)$

d. $(x + 14)(x - 1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Factoring quadratic expressions

Factor $2x^2 - x - 6$.

a. $(2x + 3)(x - 2)$

b. $(2x - 3)(x + 2)$

c. $(x + 3)(2x - 2)$

d. $(2x + 1)(x - 6)$

30 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$

31 [Intermediate] • Target: MCQ — Absolute value equations

Solve $|3x + 2| = 13$.

a. $x = \frac{11}{3}$ or $x = 5$

b. $x = 5$ or $x = -5$

c. $x = -\frac{11}{3}$ or $x = 5$

d. $x = \frac{11}{3}$ or $x = -5$

32 [Intermediate] • Target: MCQ — Linear inequalities

Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$

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33 [Intermediate] • Target: MCQ — Rational equations

Solve $\frac{x}{x-3} = 4$.

a. $x = 12$

b. $x = -4$

c. $x = 4$

d. $x = 3$

34 [Intermediate] • Target: MCQ — Rational equations

Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$

35 [Intermediate] • Target: MCQ — Linear-quadratic systems

Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

36 [Intermediate] • Target: MCQ — Solving quadratic equations by factoring

Solve $4(x + 9) = x + 36$.

a. 3

b. -2

c. 0

d. 2

37 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

Solve $2(x + 7) = 3x + 23$.

a. -11

b. -7

c. -9

d. 9

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

Solve $5(x-2)=2x-22$.

a. -2

b. -6

c. 4

d. -4

39 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $2(x+7)=x+10$.

a. 4

b. -2

c. -4

d. -6

40 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

Factor $x^2-4x-32$.

a. $(x-6)(x+4)$

b. $(x-8)(x+4)$

c. $(x+8)(x-4)$

d. $(x-8)(x-2)$

41 [Challenging] • Target: MCQ — Rational equations • Cross-Lesson: 7.01 Integration & Applications

Factor $x^2+4x-32$.

a. $(x+8)(x-4)$

b. $(x+10)(x-4)$

c. $(x-8)(x+4)$

d. $(x+8)(x+6)$

42 [Challenging] • Target: MCQ — Solving for a variable in a formula • Cross-Lesson: 8.01 Exponentials & Logarithms

Factor x^2-9.

a. $(x+3)(x-3)$

b. $(x-3)(x+3)$

c. $(x-1)(x+3)$

d. $(x-3)(x-1)$

43 [Challenging] • Target: MCQ — Linear-quadratic systems • Cross-Lesson: 9.01 Statistics & Probability

Use the quadratic formula to solve $x^2+8x+3=0$, leaving the answer in exact form.

a. $\frac{-8\pm\sqrt{56}}{2}$

b. $\frac{8\pm\sqrt{52}}{2}$

c. $\frac{-8\pm\sqrt{52}}{2}$

d. $\frac{-8\pm\sqrt{48}}{2}$

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44 [Challenging] • Target: MCQ — Word problems: setting up and solving equations • Cross-Lesson: 2.01 Functions & Transformations

Find the discriminant of $3x^2 - 4x + 9 = 0$ and state how many real roots it has.

a. 92, no real roots

b. -92, no real roots

c. -92, two distinct real roots

d. -92, one repeated real root

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Sequences & Series

Solve $|x - 9| = 11$.

a. $-2, 20$

b. $22, -2$

c. $20, 0$

d. $20, -2$

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